Ondřej Kubů

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PROFILE

Severo Ochoa postdoc at ICMAT-CSIC Madrid, now working on **mathematical problems in compute verification**. PhD in mathematical physics: main research in differential geometry and integrable systems. 10 peer-reviewed articles.

AI VERIFICATION

Verification research: *Proof of useful work for verifying AI treaties* (Oct 2026), a technical literature review with a list of open problems. Sparked by discussions with the GPAI Policy Lab verification research group, with feedback from P. Wang.

Iliad Intensive, LISA London (Nov 2026), admitted.

Compute Verification 1, LENS Academy (Sep–Oct 2026), certificate (Professional tier). Cohort course on treaty verification, from precedents and evidence to hardware mechanisms.

AGI Strategy course, BlueDot Impact (Jul 2026), certificate. Chose compute-governance verification through hardware-enabled mechanisms as the topic for the final project.

Public writing: *ARC's research agenda: solid mathematics, an unclear safety case* (Jul 2026), first in a series on AI-safety orgs for mathematicians moving into the field; and *AI in Mathematics, 2026: From Assistant to Top Researcher* (Aug 2026).

Newsletter author, *Pozastavení nad AI* (for PauseAI CZ, 2026–present).

Giving What We Can 10% Pledge. Committed to donating 10% of lifetime income to effective charities.

EMPLOYMENT

Dec 2024 – Oct 2026	**Postdoc (Severo Ochoa)**, Instituto de Ciencias Matemáticas (ICMAT), CSIC, Madrid. Haantjes algebras for separation of variables of (super)integrable systems. **Supervisor:** P. Tempesta
Sep 2024 – Nov 2024	**Research Associate (part-time)**, FNSPE, Czech Technical University in Prague
2021 – 2024	**Assistant lecturer**, Dept. of Physics, FNSPE CTU. Seminars in Theoretical Physics 2, Lie Algebras and Groups, and Geometrical Methods in Physics.

EDUCATION

2020 – 2024	**Ph. D.**, FNSPE CTU Prague, Mathematical Engineering / Mathematical Physics. **Thesis:** *Integrability and superintegrability in the presence of magnetic fields: separable and nonseparable systems.* **Supervisor:** L. Šnobl; **Co-supervisor:** A. Marchesiello. *Rector's Award for Excellent Doctoral Thesis (2025).* **Grade:** 1.00 GPA (Excellent); oral defence Grade 1.
2018 – 2020	**Master's Degree with Honors (Ing.)**, FNSPE CTU, Mathematical Physics. **Supervisor:** L. Šnobl; **Consultant:** P. Winternitz. **Grade:** A (GPA 1.31).
2015 – 2018	**Bachelor's Degree with Honors**, FNSPE CTU. **Grade:** A (GPA 1.41).

RESEARCH VISITS

ERASMUS+ research exchanges: Universidad Complutense de Madrid, Spring 2023 (P. Tempesta); Université de Montréal, Fall 2019 (P. Winternitz).

AWARDS

Rector's Award for Excellent Doctoral Thesis, CTU (2025); **Josef Hlávka Award** for the best students and graduates of public universities in Prague (2023); **Milan Odehnal Award** (2024), honorable mention, Czech Physical Society; **Dean's Award** for the 2nd best Master Thesis (2021).

SKILLS

Software: Maple, LaTeX; author of an open Maple package for computing Nijenhuis/Haantjes torsion, github.com/OndrejKubu/NijenhuisHaantjesTorsion (2026). Basics of C++; daily use of AI assistants in research.

Coursework: Doctoral courses, all graded 1 (excellent): Schrödinger operators; geometrical aspects of spectral theory; operator semigroups; classification and identification of Lie algebras; solvable models of mathematical physics.

Service: 4 peer reviews for *Journal of Physics A*.

Languages: Czech (native), English (C1–C2), German (B2/C1), French (B2), Spanish (A2).

PUBLICATIONS

Full list: ORCID 0000-0001-7463-5680.

Journal articles

O. Kubů, D. Latini. **Haantjes algebras, Zernike system and separation of variables**. *J. Phys. A* (2026), in press. DOI: 10.1088/1751-8121/aeaf25. arXiv:2605.23748

O. Kubů, P. Tempesta. **Stäckel and Eisenhart lifts, Haantjes geometry and Gravitation**. *Proc. R. Soc. A* **482** (2026) 20251103. DOI: 10.1098/rspa.2025.1103. arXiv:2509.19950

O. Kubů, D. Reyes, P. Tempesta, G. Tondo. Hamiltonian integrable systems in a magnetic field and symplectic-Haantjes geometry. *Proc. R. Soc. A* **480** (2024) 20240076

O. Kubů, A. Marchesiello, L. Šnobl. Integrable systems with velocity-dependent potentials: generalized parabolic cylindrical case. *J. Phys. A* **57** (2024) 235203

O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems in magnetic fields: the generalized cylindrical and spherical cases. *Ann. Phys.* **451** (2023) 169264

O. Kubů, L. Šnobl. Cylindrical first-order superintegrability with complex magnetic fields. *J. Math. Phys.* **64** (2023) 062101. *(Non-self-adjoint Hamiltonians: pseudo-Hermiticity without quasi-Hermiticity.)*

M. F. Hoque, O. Kubů, A. Marchesiello, L. Šnobl. New classes of quadratically integrable systems with velocity dependent potentials: non-subgroup type cases. *Eur. Phys. J. Plus* **138** (2023) 845

O. Kubů, A. Marchesiello, L. Šnobl. Superintegrability of separable systems with magnetic field: the cylindrical case. *J. Phys. A* **54** (2021) 425204

S. Bertrand, O. Kubů, L. Šnobl. On superintegrability of 3D axially-symmetric non-subgroup-type systems with magnetic fields. *J. Phys. A* **54** (2021) 015201

O. Kubů, L. Šnobl. Superintegrability and time-dependent integrals. *Archivum Math.* **55** (2019) 309–318

SELECTED TALKS

XXXIV International Fall Workshop on Geometry and Physics, Tarragona (Sep 2026); *36th International Colloquium on Group Theoretical Methods in Physics* (GROUP36), Valladolid (Jul 2026); **Invited**, *Geometry and Integrable Systems*, La Trobe University, Melbourne (Feb 2026); **Invited**, *Superintegrability, Exact-Solvability, and Representation Theory* (Nov 2024). Full list on the website.